

# 4 File permissions in Linux

## 4.1 Permissions and Restrictions

In the section on managing files and folders, we talked about how everything in Linux, be it a directory, an actual file, or even a device are just files to the operating system and all of these have permissions that allow or restrict others from viewing, modifying or executing them. If the file is of the type Directory then it has different permissions and restrictions than say, files and device nodes and each of these have access restrictions with permissions and user restrictions associated with owner/group association. However, the super user **root** has the ability to access any file on the system. These permissions are referred to as bits. If the owner read & execute bits are on, then the corresponding permissions are **-r-x-----**

There are three types of access restrictions:

| Permission | Action    | chmod option |
|------------|-----------|--------------|
| read       | (view)    | r or 4       |
| write      | (edit)    | w or 2       |
| execute    | (execute) | x or 1       |

There are also three types of user restrictions:

| User  | ls output |
|-------|-----------|
| owner | -rwx----- |
| group | -rwx----- |
| other | -rwx----- |

These restrictions, however are not inheritable, as in the case where restrictions are set for the owner's group or 'everyone', the file owner will still be unaffected with full rights.

We mentioned previously that directories have directory permissions, which are different from file and device node permissions. These are:

| Permission | Action                                                                    | chmod option |
|------------|---------------------------------------------------------------------------|--------------|
| read       | restricts or allows viewing the directories contents, i.e. ls command     | r or 4       |
| write      | restricts or allows creating new files or deleting files in the directory | w or 2       |
| execute    | restricts or allows changing into the directory, i.e. cd command          | x or 1       |

Example:

If a file is owned by the user root and belongs to the root group, the